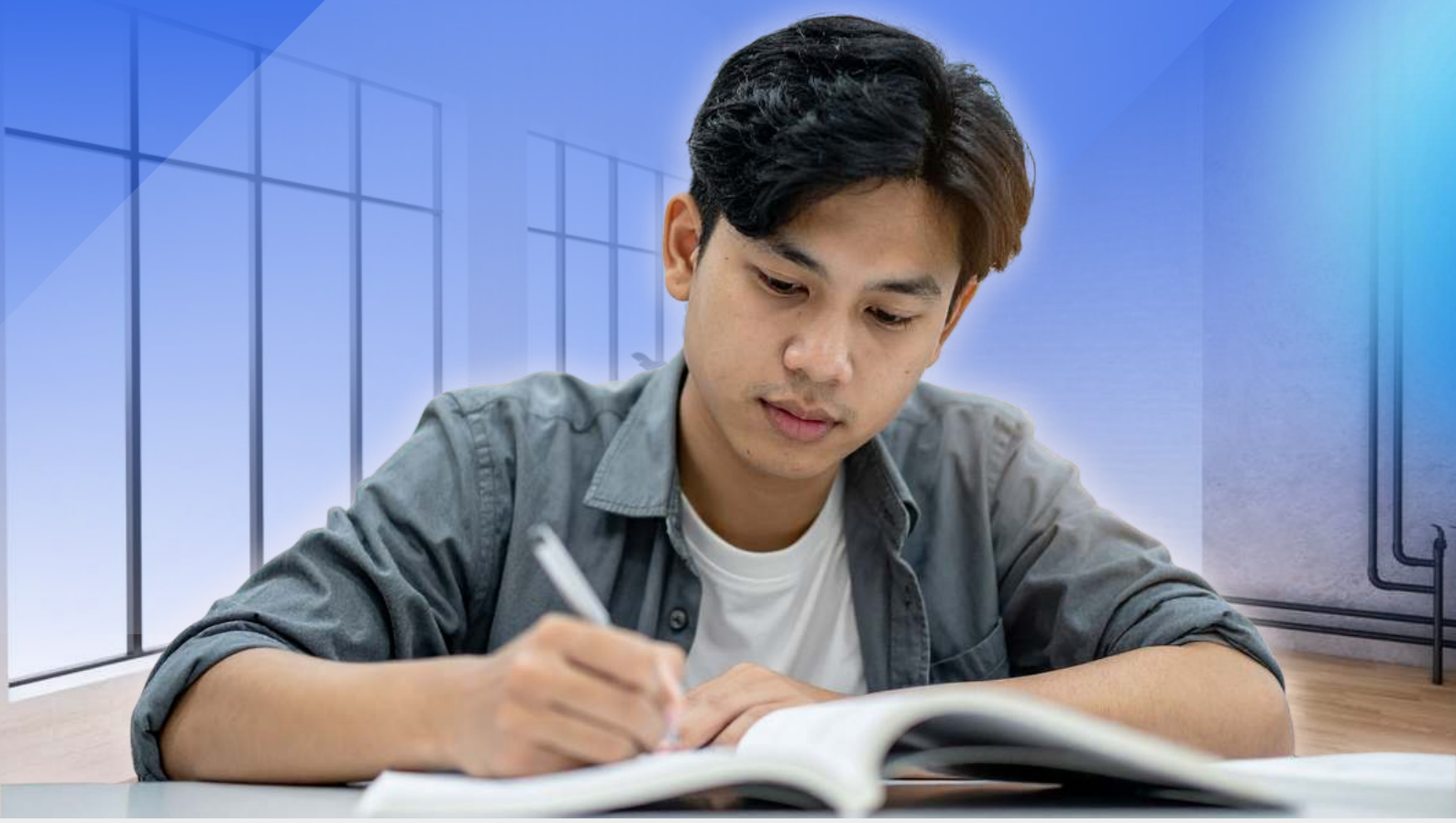
Lenterago

TRY OUT #6

SAT-Math

61006

54 QUESTIONS



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MODUL 1 & 2

22 QUESTIONS

DIRECTIONS

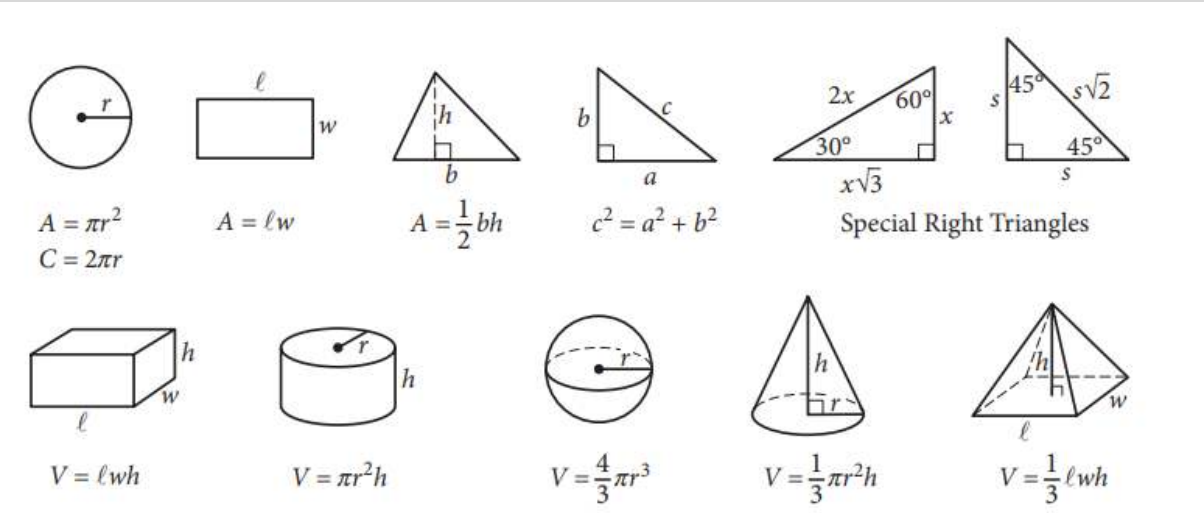
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function *f* is the set of all real numbers *x* for which *f(x)* is a real number

REFERENCE



The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2p.

The sum of the measures in degrees of the angles of a triangle is 180.

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Module 1 & 2

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find more than one correct answer, write and circle only one answer.
- Your answer can be up to 5 characters for a positive answer and up to 6 characters (including the negative sign) for a negative answer, but no more.
- If your answer is a fraction that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a decimal that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a mixed number (such as 3 1/2), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include symbols such as a percent sign, comma, or dollar sign in your circled answer.

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NO	MATERI	SUB MATERI	
1	Linear Equation		
	If $9(4-3x)+2=8(4-3x)+18$, what is the value of $4-3x$? A. -16 B. -4 C. 4 D. 16		
2	Imaginary Number	Operation	
	If $a+bi=(1+2i)(3-4i)$ where a and b constant and $i=\sqrt{-1}$ what is the value of $a+b$?		
3	Function & Exponent		
	The number of bacteria in a liquid medium doubles every day. There are 44,000 bacteria in the liquid medium at the start of an observation. Which of the following represents the number of bacteria, y, in the liquid medium t days after the start of the observation? A. $y=\frac{1}{2}(44000)^t$ B. $y=2(44000)^t$ C. $y=44000(\frac{1}{2})^t$ D. $y=44000(2)^t$		
4	Imaginary Number	Operation	
	If $i^m=-i$ which of following CANNOT be the value of m? A. 15 B. 18 C. 19 D. 27		
5	Exponents & Radicals	Exponents & Radicals Equation	
	If $m^3=\sqrt{\sqrt{n}}$, where $n>0$, what is the value of m in terms of n? A. $n^{\frac{1}{12}}$ B. $n^{\frac{1}{7}}$ C. $n^{\frac{7}{12}}$ D. $n^{\frac{3}{4}}$		
6	Function		
	To edit a video, Michael charges \$50 for the first 2 hours and \$20 per hour after the first 2 hours. Which of the following express the amount C, in dollars, Michael charges if it take him x hours to edit the video, where $x>2$? A. $C=20x+10$ B. $C=20x+25$ C. $C=20x+50$ D. $C=20x+100$		

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7	Ratio, Proportion	Ratio	
	On Thursday, 240 adults and children attended a show. The ratio of adults to children was 5 to 1. How many children attended the show? A. 40 B. 48 C. 192 D. 200		
8	Statistics	Median	
	The median of numbers $x, 10$ and 12 is 12. Which of the following CANNOT be the value of x ? A. 8 B. 12 C. 16 D. 20		
9	Linear Equation	Word Problem	
	A bus is traveling at a constant speed along a straight portion of road. The equation d = 30t gives the distance d, in feet from a road marker, that the bus will be t seconds after passing the marker. How many feet from the marker will the bus be 2 seconds after passing the marker? A. 30 B. 32 C. 60 D. 90		
10	Probability	Probability	
	For a particular machine that produces beads, 29 out of every 100 beads it produces have a defect. A bead produced by the machine will be selected at random. What is the probability of selecting a bead that has a defect? A. $\frac{1}{2900}$ B. $\frac{1}{29}$ C. $\frac{29}{100}$ D. $\frac{29}{10}$		
11	Function	Linear Function	
	$f(x) = 4x + b$ For the linear function f, b is a constant and $f(7) = 28$. What is the value of b? A. 0 B. 1 C. 4 D. 7		
12	Linear Equation	Line Equation	
	What is the equation of the line that passes through the point (0, 5) and is parallel to the graph of $y = 7x + 4$ in the xy-plane? A. $y = 5x$ B. $y = 7x + 5$ C. $y = 7x$ D. $y = 5x + 7$		

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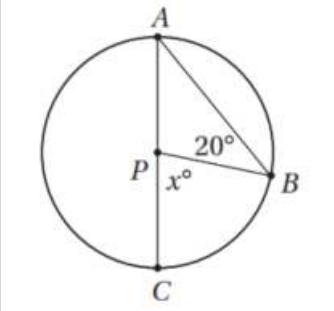
13	Function and Percent	Function and Percent	
	For $x > 0$ the function f is defined as follows : $f(x)$ equals 201% of x Which of the following could describe this function ? A. Decreasing exponential B. Decreasing linear C. Increasing exponential D. Increasing linear		
14	Linear Equation	System of Linear Equation	
	$4x - 9y = 9y + 5$ $hy = 2 + 4x$ In the given system of equation, h is a constant. . If the system has no solution, what is the value of h ? A. -9 B. 0 C. 9 D. 18		
15	Quadratic		
	In the expression $x^2 + kx + 18$, k is a negative integer. Which of the following is NOT possible value of k? A. -19 B. -11 C. -9 D. -7		
16	Triangle	Pythagoras Theorem	
	A right triangle has legs with lengths of 24 centimeters and 21 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d ?		
17	Circle and Angle	Circle and Angle	
	 In the figure above, P is the center of a circle and AC is its diameter. What is the value of x? A. 60 B. 50 C. 40 D. 30		
18	Triangles and Trigonometry	Triangles and Trigonometry	

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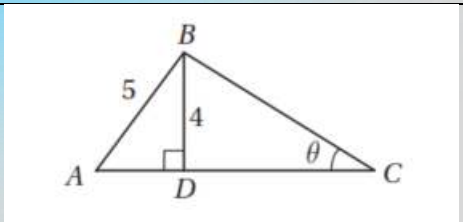
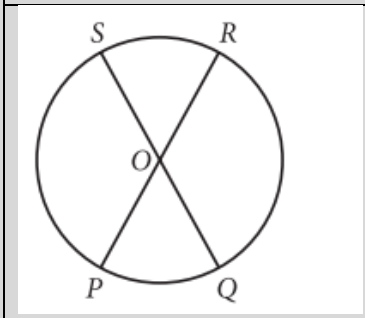


In figure above, triangle ABC has an area of 19. What is the value of $\tan\theta$

19	Percent	Percent	
	13 is p% of 35. What is the value of p ?		
20	Circle	Arc of circle	



The circle shown has center O, circumference 144π, and diameters $\overline{PQ}$ and $\overline{QS}$. The length of arc PS is twice the length of arc PQ . What is the length of arc QR ?

A. 24π
B. 48π
C. 72π
D. 96π

21	Function	Minimum / maximum value	

$f(x) = 4x^2 + 64x + 262$

The function g is defined by $g(x) = f(x+5)$. For what value of x does $g(x)$ reach its minimum?

A. -13
B. -8
C. -5
D. -3

22	Three-Dimensional Figures	Area and Volume of Rectangular Prism	

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 58 square centimeters (cm^2) and the surface area of right rectangular prism Y is 1,450 cm^2. The volume of right rectangular prism Y is 1,250 cubic centimeters (cm^3). What is the sum of the volumes, in cm^3, of right rectangular prism X and right rectangular prism Y?

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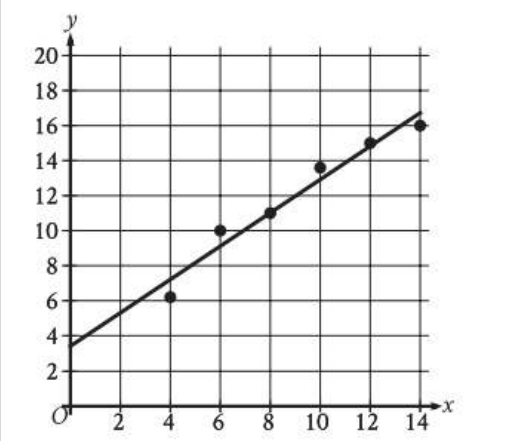
23	Inequality	Word problem into math equation	
	The total cost, in dollars, to rent a surfboard consists of a \$25 service fee and a \$10 per hour rental fee. A person rents a surfboard for t hours and intends to spend a maximum of \$75 to rent the surfboard. Which inequality represents this situation? A. $10t \leq 75$ B. $10 + 25t \leq 75$ C. $10t + 25 \geq 75$ D. $25 + 10t \leq 75$		
24	Linear Equation	Word problem	
	Vivian bought party hats and cupcakes for \$71. Each package of party hats cost \$3, and each cupcake cost \$1. If Vivian bought 10 packages of party hats, how many cupcakes did she buy?		
25	Rate & Proportion	Rate & Proportion	
	The number of squirrels in a 131 square mile area is estimated to be 2358. What is the estimated population density, in squirrels per square mile of this area? A. 18 B. 131 C. 149 D. 2376		
26	Scatterplot	Equation of the scatterplot	
	The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.  Which of the following equations best represents the line of best fit shown? A. $y = x + 3.4$ B. $y = x - 3.4$ C. $y = -x + 3.4$ D. $y = -x - 3.4$		
27	Trigonometry	Value of Trigonometry	
	In right triangle RST , the sum of the measures of angle R and angle S is 90 degree. The value of $\sin(R) = \frac{\sqrt{15}}{4}$. What is the value of $\cos(S)$? A. $\frac{\sqrt{15}}{15}$ B. $\frac{\sqrt{15}}{4}$ C. $\frac{4\sqrt{15}}{15}$ D. $\sqrt{15}$		

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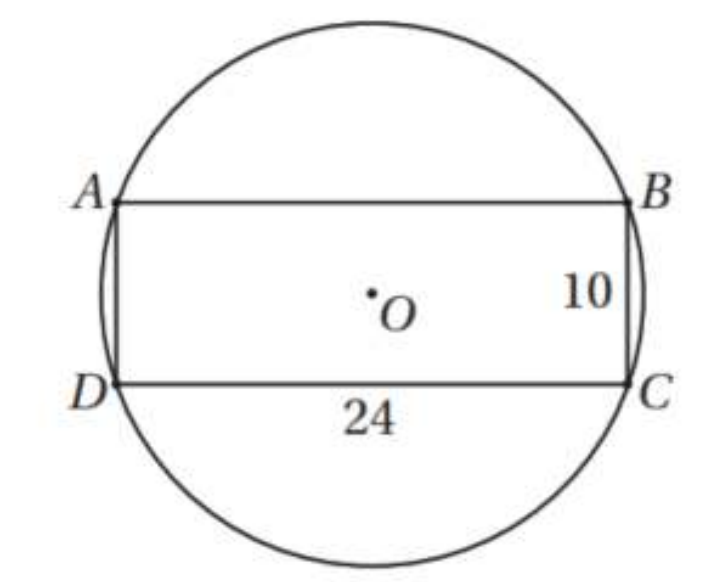
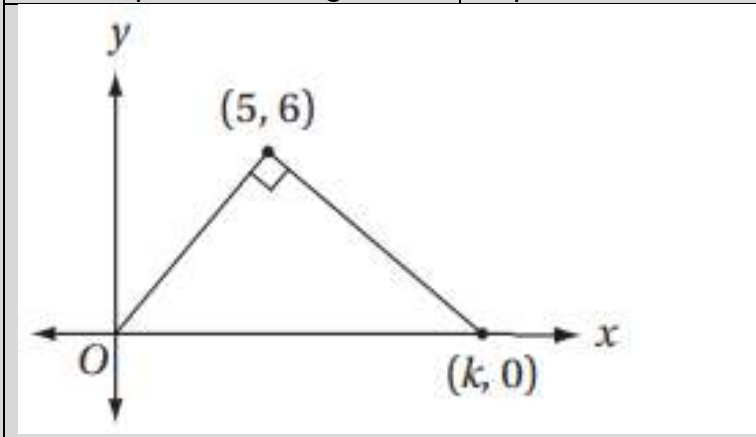
28	Circles	Area of Circles + Pythagoras Theorem



The figure above show that rectangle ABCD is inscribed in the circle with center O. What is the area of circle?

A. 26π
B. 121π
C. 144π
D. 169π

29	Linear Equation + Triangles	Slope of 2 lines



The figure above show a tight triangle with vertices at the origin , $(5,6)$ and $(k,0)$. What is the value of k ?

A. $\frac{19}{3}$
B. $\frac{58}{5}$
C. $\frac{26}{3}$
D. $\frac{61}{5}$

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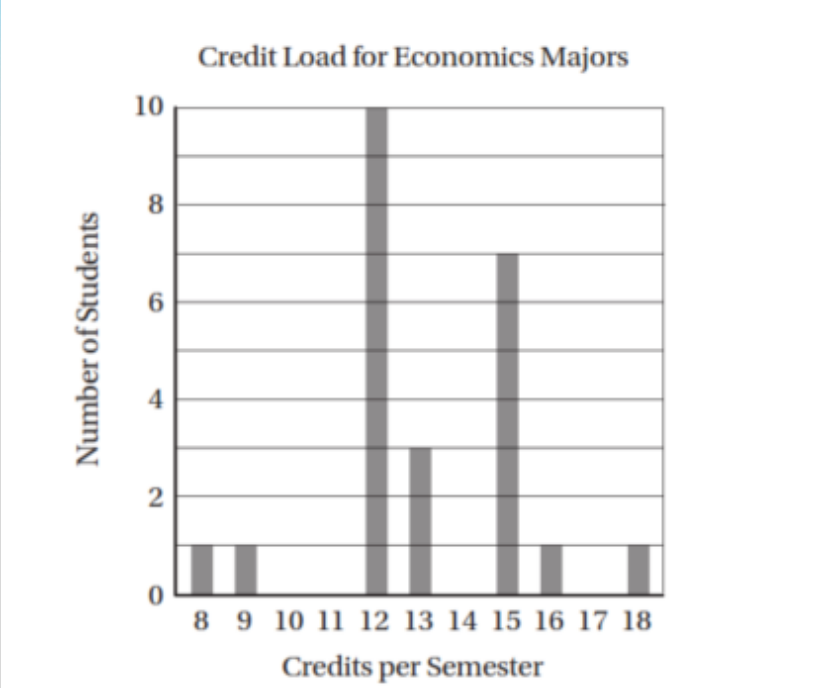
30	Table and Statistics	Table, Statistics and Percentage	

Question number 8 and 10 are based on the graph below



A university surveyed 24 economic major and asked them how many credit they received the previous semester. The result are represented in the graph above . What percentage of these student received 15 or more credits that semester?

A. 29%
B. $33\frac{1}{3}\%$
C. $37\frac{1}{2}\%$
D. 54%

31	Table and Statistics	Median	

What is the median number of credits these students received the previous semester?

A. 10.5
B. 11.5
C. 12
D. 12.5

32	Rates, Proportions, and Percents	Proportions	

A website received 2100 visitor in April from both subscribers and nonsubscribers. If the ratio of subscriber to nonsubscribers among this group was 2:5. How many nonsubscriber visited the site in April than subscribers?

A. 126
B. 630
C. 900
D. 1260

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33 | Probability | Probability

	Live east of the river	Live west of the river	Total
Less than 40 years old	17	11	28
At least 40 years old	18	89	107
Total	35	100	135

The table summarizes members of a local organization by age and whether they live east or west of the river. If a member of the organization is selected at random, what is the probability that the selected member is at least 40 years old?

A. $\frac{28}{135}$
B. $\frac{35}{135}$
C. $\frac{100}{135}$
D. $\frac{107}{135}$

34 | Function

x	$f(x)$
-4	0
$-\frac{19}{5}$	1
$-\frac{18}{5}$	2

For linear function f, the table shows three value of x and their corresponding value of $f(x)$. If $h(x) = f(x) - 13$, which equation define h?

A. $h(x) = 5x - 4$
B. $h(x) = 5x + 7$
C. $h(x) = 5x + 9$
D. $h(x) = 5x + 20$

35 | Quadratic Equation | Type of Solution

$x(kx - 56) = -16$
In the given equation, k is an integer constant. If the equation has no real solution, what is the least possible value of k ?

36 | Exponential Function | Word Problem

The function $f(t) = 60000(2)^{\frac{t}{410}}$ give the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

0 8 2 1 7 0 2 7 6 0 7 6

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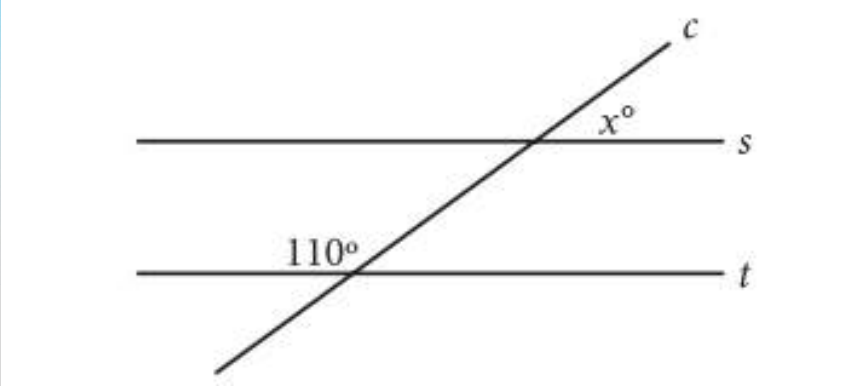
37	Line and Angle	The angle in a line intersect with 2 parallel line	



In the figure shown, line c intersects parallel lines s and t. What is the value of x ?

38	Polynomial	Operation of Polynomial	

$$3x^3 - 2x^2 + 5$$
$$5x^2 + x - 10$$

Which of the following is the sum of two polynomials shown above?

A. $8x^3 - 2x - 5$
B. $3x^3 - x^2 - 5$
C. $3x^3 + 3x^2 + x - 5$
D. $8x^5 - x^3 - 5$

39	Quadratic Function	Function and Quadratic	

Which of the following function when graphed in the xy-plane has exactly one negative x −intercept and one negative y −intercept?

A. $y = -x^2 - 6x - 9$
B. $y = -x^2 + 6x - 9$
C. $y = x^2 + 6x + 9$
D. $y = x^2 - 6x + 9$

40	Linear Equation	Word problem	

A bowl contains 20 ounces of water. When the bowl is uncovered, the amount of water in the bowl decreases by 1 ounce every 4 days. If 9 ounces of water remain in this bowl, for how many days has it been uncovered?

A. 3
B. 7
C. 36
D. 44

41	System Equation	Equation	

If $\frac{x}{y} = 4$ and $\frac{24x}{ny} = 4$, what is the value of n ?

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42	Inequality	Inequality result in table

$$y > 13x - 18$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	21
5	47
8	86

B.

x	y
3	26
5	42
8	86

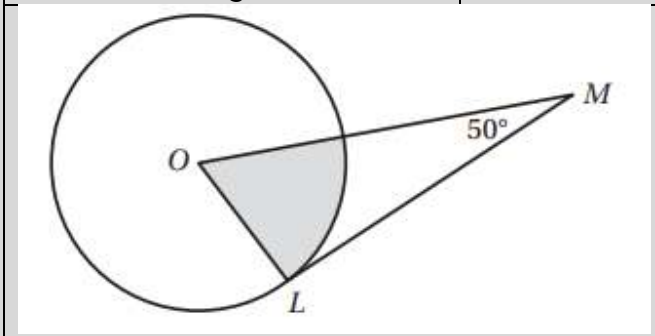
C.

x	y
3	16
5	42
8	81

D.

x	y
3	26
5	52
8	91

43	Circle and Triangle	Area of sector



In the figure above the circle has a center O, line segment LM is tangent to the circle at point L and angle LMO has measure of $50°$. If the area of the circle is 108 square centimeters, what is the area, in square centimeters, of the shaded sector?

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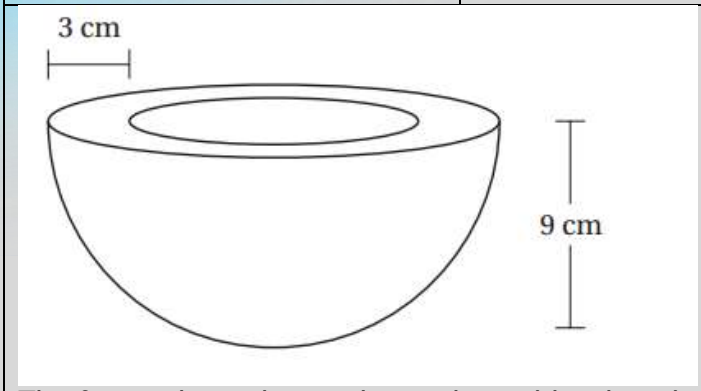
44	Three-Dimensional Figures	Volume of spere	



The figure above shows a hemispherical bowl made of glass. The bowl is 9 centimeters high and the glass 3 centimeter thick. A second bowl is to be constructed to scale with the original bowl, but in one-half the height and diameter. The smaller bowl can hold a maximum ok $k\pi$ cubic centimeters of water. What is the value of k?

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